

数论基础课程作业

2026-07-04

第一章

Exercise Sheet 1

Exercise 1. $\pi(x) \geq \log \log x$.

Solution. pf: $p_{n+1} \leq p_1 \cdots p_n + 1$, 归纳, $p_n \leq 2^{2^{n-1}} \leq e^{e^{n-1}}$, 取对数。

Exercise 2. (a) $\tau(n^2) = \sum_{d|n} \mu(d) \tau(n/d)^2$; (b) $\sum_{d^k|n} \mu(d) = 1$ 无 k 次方因子; (c) $\mu(n) = \sum_{1 \leq a \leq n, (a,n)=1} e^{2\pi i a/n}$.

Solution. (a) pf: $\tau(\cdot)^2 = \mu * \tau(\cdot)^2$, 只需 $\tau(\cdot)^2 = \tau(\cdot)^2 * u$, 显。(b) pf: 最大 k 次方因子设为 $p_1^{k\alpha_1} \cdots p_m^{k\alpha_m}$, 若为 1, 则 LHS=1, 否则为 $\sum_{d|p_1 \cdots p_m} \mu(d) = 0$ 。(c) pf: 先证 RHS 可乘, 因为 m, n 互素时, 在模 mn 意义下 $\{am + bn \mid a \in (\mathbb{Z}/n)^\times, b \in (\mathbb{Z}/m)^\times\} = (\mathbb{Z}/mn)^\times$, 再说 p^α 时 LHS=RHS (用 p 进制)。

Exercise 3. (a) $\sum_{d|n} \mu(d)^2 = 2^{\omega(n)}$; (b) $\sum_{d|n} \mu(d) \tau(d) = (-1)^{\omega(n)}$.

Solution. (a) pf: 显。(b) pf: LHS = $\sum_{k=0}^{\omega(n)} \binom{\omega(n)}{k} (-1)^k 2^k = (1-2)^{\omega(n)} = (-1)^{\omega(n)}$ 。

Exercise 4. f 可乘, 则 f^{-1} 完全可乘 iff $f(p^\ell) = 0, \ell \geq 2$.

Solution. pf: 已知可乘则完全可乘等价于 $f^{-1} = \mu f$ 故显。

Exercise 5. $\{f : f * f = u\}$ 恰有两个, 是否可乘?

Solution. pf: f 满足 $f(1) = \pm 1$, 递推 $f(n) = \frac{1}{2f(1)}(1 - \sum_{d|n, d \neq 1, n} f(d)f(n/d))$, 该递推确定, 故恰有两个。若 $f(1) = -1$ 不可乘; $f(1) = 1$ 可乘 (用反证法找最小的 mn)。

Exercise 6. $(f * g)^{-1} = f^{-1} * g^{-1}$.

Solution. 显。

Exercise 7. $\text{supp } \Lambda_k \subseteq \{n : \omega(n) \leq k\}$; $0 \leq \Lambda_k(n) \leq (\log n)^k$.

Solution. ① pf: $\Lambda_k = \Lambda_{k-1} \log + \Lambda_{k-1} * \Lambda$, 故显。② $\Lambda \geq 0$, 是因为递推式, “ $\leq$ ” 是因为 $\log^k = \Lambda_k * u \Rightarrow \Lambda_k(n) = \log^k n - \sum_{d|n, d < n} \Lambda_k(d) \leq \log^k n$ 。

第二章

Exercise Sheet 2

Exercise 1. 求 $f \in A$ s.t. $\frac{1}{\varphi(n)} = \sum_{d|n} \frac{1}{d} f(n/d)$.

Solution. pf: $a(n) = 1/n$ 有 $a^{-1}(n) = \mu(n)/n$, 故 $f = \frac{1}{\varphi} * a^{-1}$ 可乘, $f(p) = \frac{1}{p(p-1)}$, $f(p^\alpha) = 0$ ($\alpha \geq 2$), 故 $f(n) = \mu(n)^2 \prod_{p|n} \frac{1}{p(p-1)} = \frac{\mu(n)^2}{n\varphi(n)}$.

Exercise 2. $\sum_{n \leq x} \frac{1}{\varphi(n)} = C_1 \log x + C_2 + O(\log x/x)$.

Solution. pf: $\sum_{n \leq x} \frac{1}{\varphi(n)} = \sum_{n \leq x} a * f(n) = \sum_{d \leq x} f(d) \sum_{m \leq x/d} \frac{1}{m} = \sum_{d \leq x} f(d) (\log \frac{x}{d} + \gamma + O(d/x)) := \log x \sum_{d \geq 1} f(d) + (-\sum_{d \geq 1} f(d) \log d + \gamma \sum_{d \geq 1} f(d)) - \log x \sum_{d > x} f(d) - (-\sum_{d > x} f(d) \log d + \gamma \sum_{d > x} f(d)) + O(\sum_{d \leq x} f(d)d/x)$. 由以下估计可证原题: ① $\sum_{d \geq 1} f(d)$, $\sum_{d \geq 1} f(d) \log d$ 收敛. pf: $\varphi(n) \gg n/\log \log n$ (由此也可知 $\sum_{n \leq x} 1/\varphi(n) = O(\log x)$). ② $\sum_{d \leq x} f(d)d = O(\log x)$. pf: $|\sum_{d \leq x} f(d)d| = |\sum_{d \leq x} \mu(d)^2/\varphi(d)| \leq |\sum_{d \leq x} 1/\varphi(d)| = O(\log x)$. ③ $\sum_{d > x} f(d) = O(1/x)$. pf: $f(d) = d^{-2} \mu(d)^2 d/\varphi(d)$, 记为 $d^{-2} b(d)$, 由 2-1, $\frac{n}{\varphi(n)} = n a * f(n) = n \sum_{d|n} \frac{d \mu^2(d)}{n d \varphi(d)} = \sum_{d|n} \frac{\mu^2(d)}{\varphi(d)}$, 即 $|b(n)| \leq n/\varphi(n) = \sum_{d|n} \mu^2(d)/\varphi(d)$, 那 $|\sum_{n \leq x} b(n)| \leq \sum_{n \leq x} \sum_{d|n} \mu^2(d)/\varphi(d) = \sum_{d \leq x} \frac{\mu^2(d)}{\varphi(d)} [x/d] \leq x \sum_{d \leq x} \frac{\mu^2(d)}{d \varphi(d)} = x \sum_{d \leq x} f(d) = O(x)$, 即 $\sum_{d > x} f(d) = \lim_{N \rightarrow \infty} \sum_{x < d \leq N} \frac{b(d)}{d^2} = \lim_{N \rightarrow \infty} \int_x^N \frac{1}{\xi^2} d \sum_{x < d \leq \xi} b(d) = \lim_{N \rightarrow \infty} [O(1/N) - 2 \int_x^N \xi^{-3} \sum_{x < d \leq \xi} b(d) d\xi] = 2 \int_x^\infty \xi^{-3} O(\xi) d\xi = O(1/x)$. ④ $\sum_{d > x} f(d) \log d = O(\log x/x)$. pf: LHS = $\sum_{d > x} b(d) \log d d^{-2} = \lim_{N \rightarrow \infty} \sum_{x < d \leq N} b(d) \frac{\log d}{d^2} = \lim_{N \rightarrow \infty} \int_x^N \frac{\log \xi}{\xi^2} d \sum_{x < d \leq \xi} b(d) = O(\log N/N) - \int_x^N \frac{1 - \log \xi}{\xi^3} \sum_{x < d \leq \xi} b(d) d\xi = 2 \int_x^\infty \frac{\log \xi - 1}{\xi^3} O(\xi) d\xi = O(\log x/x)$ 注意 $\log x + 1/x$ 导数是 $-\log x/x^2$, 故得证

Exercise 3. 对 $x > 3$, $\sum_{1 < n \leq x} \frac{1}{n \log n} = \log \log x + C_1 + O(1/(x \log x))$.

Solution. pf: Euler–Maclaurin, $\text{LHS} = \int_2^x \frac{d\xi}{\xi \log \xi} + \int_2^x (\{\xi\} - \frac{1}{2}) (\frac{1}{\xi \log \xi})' d\xi + (\{x\} - \frac{1}{2}) \frac{1}{x \log x} - (\{2\} - \frac{1}{2}) \frac{1}{2 \log 2} = \{\log \log x\} - \{\log \log 2\} + \frac{1}{4 \log 2} + \int_2^\infty [\{\xi\} - \frac{1}{2}] (\frac{1}{\xi \log \xi})' d\xi - \{\int_x^\infty [\{\xi\} - \frac{1}{2}] (\frac{1}{\xi \log \xi})' d\xi + (\{x\} - \frac{1}{2}) \frac{1}{x \log x}\}$, 即为所求。

Exercise 4. $r_2(n) := \#\{(a, b) \in \mathbb{Z}^2 \mid a^2 + b^2 = n\}$, 求证 $\sum_{n \leq x} r_2(n) = \pi x + O(\sqrt{x})$.

Solution. pf: $\sum_{n \leq x} r_2(n) = 4 \sum_{0 \leq a \leq \sqrt{x}} \sum_{0 < b < \sqrt{x-a^2}} 1 + 1 = 4 \sum_{0 \leq a \leq \sqrt{x}} [\sqrt{x-a^2}] + 1 = 4 \sum_{0 \leq a \leq \sqrt{x}} \sqrt{x-a^2} + O(\sqrt{x})$. 关于 a 单调, $\sum_{0 \leq a \leq \sqrt{x}} \sqrt{x-a^2} = \int_0^{\sqrt{x}} \sqrt{x-a^2} da + O(\sqrt{x}) = -\frac{x \arccos(a/\sqrt{x})}{2} \Big|_0^{\sqrt{x}} + \frac{a\sqrt{x-a^2}}{2} \Big|_0^{\sqrt{x}} + O(\sqrt{x}) = \frac{\pi}{4}x + O(\sqrt{x})$, 且 $a=0$ 值也为 $O(\sqrt{x})$.

Exercise 5. $Q_2(x) = \#\{n \leq x \mid n \text{ 无平方因子}\}$, 则 $Q_2(x) = C_2 x + O(\sqrt{x})$ ($C_2 = 6/\pi^2$).

Solution. pf: $Q_2(x) = \sum_{n \leq x} \mu^2(n) = \sum_{n \leq x} \sum_{d^2 | n} \mu(d) = \sum_{d \leq \sqrt{x}} [x/d^2] \mu(d) = \sum_{d \leq \sqrt{x}} \frac{x \mu(d)}{d^2} + O(\sqrt{x}) = x \sum_{d \geq 1} \frac{\mu(d)}{d^2} + x \sum_{d > \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x}) = \frac{1}{\zeta(2)} x + O(x \int_{\sqrt{x}}^\infty \xi^{-2} d\xi) + O(\sqrt{x}) = \frac{1}{\zeta(2)} x + O(\sqrt{x})$.

Exercise 6. $\forall \varepsilon > 0, \exists C_\varepsilon > 0, \tau(n) < C_\varepsilon n^\varepsilon$.

Solution. pf: 选 P s.t. $\forall p > P, \forall \alpha \in \mathbb{N}_{\geq 1}, \alpha + 1 < p^{\varepsilon \alpha}$, 再取 $C_\varepsilon = \prod_{p < P} \sup_{\alpha \geq 1} (\alpha + 1)/p^{\alpha \varepsilon} < \infty$, 则 $\tau(\prod p_i^{\alpha_i}) = \prod (\alpha_i + 1) \leq C_\varepsilon \prod p_i^{\varepsilon \alpha_i}$.

第三章

Exercise Sheet 3

Exercise 1. 若 $\sum_{p \leq x} f(p) \log p = (ax + b) \log x + cx + O(1)$ ($x \geq 2$), 则 $\exists B = B(f)$, $\sum_{p \leq x} f(p) = ax + (a + c)(\frac{x}{\log x} + \int_2^x \frac{dt}{\log^2 t}) + b \log \log x + B + O(\frac{1}{\log x})$.

Solution. pf: $A(x) := \sum_{p \leq x} f(p) \log p$, LHS = $\sum_{2 \leq n \leq x} f(n) \log n \delta_p(n) \frac{1}{\log n} \stackrel{\text{Abel}}{=} \frac{A(x)}{\log x} + \int_{3/2}^x A(\xi) \frac{1}{\xi \log^2 \xi} d\xi = [\text{条件}] ax + b + c \frac{x}{\log x} + O(\frac{1}{\log x}) + \int_{3/2}^x (\frac{a\xi+b}{\xi \log \xi} + \frac{c}{\log^2 \xi} + \frac{E(\xi)}{\xi \log^2 \xi}) d\xi$, 其中 $E(\xi) \ll 1$, 注意 $\frac{1}{x \log x} = (\log \log x)'$ 且 $a \int_{3/2}^x \frac{d\xi}{\log \xi} = a \frac{\xi}{\log \xi} \Big|_{3/2}^x + a \int_{3/2}^x \frac{d\xi}{\log^2 \xi}$, 积分下限可调整后放入 B 中。

Exercise 2. 证明对 $n \geq 16$, $\sum_{p|n} \frac{1}{p} \ll \log \log \log n$.

Solution. pf: $\sum_{p|n} \frac{1}{p} = \sum_{p|n, p \leq \log n} \frac{1}{p} + \sum_{p|n, p > \log n} \frac{1}{p} \ll (\text{by Mertens}) \log \log \log n + \sum_{p > \log n} \frac{1}{p}$. 由于 $(\log n)^{\log n / \log \log n} = n$ 从而 $> \log n$ 素数个数不超过 $\frac{\log n}{\log \log n}$, 于是 $\sum_{p > \log n, p|n} \frac{1}{p} \leq \frac{1}{\log n} \frac{\log n}{\log \log n} \ll \log \log \log n$.

Exercise 3. 证明 $\sum_p \frac{1}{p(\log \log p)^\alpha}$, $\alpha > 1$ 收敛, $\alpha \leq 1$ 发散.

Solution. pf: 忽略前面的项, $= \lim_{N \rightarrow \infty} S_N$, 其中 $S_N = \sum_{5 < n \leq N} \delta_p(n) \frac{1}{n} \frac{1}{(\log \log n)^\alpha} = \sum_{5 < p \leq N} \frac{1}{p} \frac{1}{(\log \log N)^\alpha} + \int_5^N (\sum_{5 < p \leq \xi} \frac{1}{p}) \frac{\alpha}{(\log \log \xi)^{\alpha+1}} \frac{d\xi}{\xi \log \xi}$, 而 $\sum_{p \leq x} \frac{1}{p} \sim \log \log x$ 且 $\int_5^\infty \frac{dx}{x \log x (\log \log x)^\alpha}$ 在 $\alpha > 1$ 收敛, $\alpha \leq 1$ 发散 (用换元 $u = \log \log x$, $\frac{dx}{x \log x} = du$) 即证。

Exercise 4. 设 $S(x) = \#\{n \leq x \mid p \mid n \Rightarrow p \leq \sqrt{n}\}$, 证明 $S(x) = (1 - \log 2)x + O(x/\log x)$.

Solution. pf: $S(x) = [x] - \sum_{2 \leq p \leq x} \sum_{m \leq x/p, m < p} 1 = [x] - \sum_{2 \leq p \leq \sqrt{x}} \sum_{m < p} 1 - \sum_{\sqrt{x} < p \leq x} \sum_{m \leq x/p} 1 = [x] - \sum_{2 \leq p \leq \sqrt{x}} (p-1) - \sum_{\sqrt{x} < p \leq x} [\frac{x}{p}]$, 而 $[x] = x + O(1)$,

$\sum_{2 \leq p \leq \sqrt{x}} (p-1) \leq \sqrt{x} \pi(\sqrt{x}) = O(x/\log x)$, $\sum_{\sqrt{x} < p \leq x} \lfloor \frac{x}{p} \rfloor = x \sum_{\sqrt{x} < p \leq x} \frac{1}{p} + O(\pi(x)) = x[(\log \log x + C + O(\frac{1}{\log x})) - (\log \log \sqrt{x} + C + O(\frac{1}{\log \sqrt{x}}))] + O(x/\log x) = x \log 2 + O(x/\log x)$, 代回即得。

Exercise 5. 若 $\sum_{n \leq x} \frac{\mu(n)}{n} = o(1)$, 则 PNT.

Solution. pf:

$$\begin{aligned} \sum_{1/2 < n \leq x} \mu(n) &= \sum_{1/2 < n \leq x} \frac{\mu(n)}{n} n \\ &\stackrel{\text{Abel}}{=} x \sum_{1/2 < n \leq x} \frac{\mu(n)}{n} - \int_{1/2}^x \sum_{1/2 < n \leq \xi} \frac{\mu(n)}{n} d\xi \\ &= o(x) + \int_{1/2}^{\sqrt{x}} M d\xi + \int_{\sqrt{x}}^x o(1) d\xi = o(x). \end{aligned}$$

故 PNT。

Exercise 6. $L(N) = \text{lcm}\{1, 2, \dots, N\}$, 求证 $\text{PNT} \Leftrightarrow \lim_{N \rightarrow \infty} L(N)^{1/N} = e$.

Solution. pf: $\log L(N) = \sum_{p^\alpha \leq N} \log p = \psi(N)$, 故 $L(N)^{1/N} \rightarrow e$ iff $\frac{\psi(N)}{N} \rightarrow 1$ iff $\psi(x) \sim x$ iff PNT。

Exercise 7. $\forall n > 1, \exists p$ s.t. $p \parallel n!$.

Solution. pf: Bertrand's 假设, 取 $\frac{n}{2} < p < n$, 则 $2p > n$, 之后显然。

第四章

Exercise Sheet 4

Exercise 1. TFAE: (a) $f(n) \ll n^A$ ($A > 0$); (b) $\sigma_a < +\infty$.

Solution. pf: ① (a) 蕴含 $|f(n)| \leq Cn^A$, 故 $|L_f(s)| \leq \sum_{n \geq 1} \frac{C}{n^{\sigma-A}}$, 即 $\text{Re}(s) > A + 1$ 时 $L_f(s)$ 绝对收敛, 那 $\sigma_a < +\infty$ 。② (b) 蕴含 $\exists c > 0$, $\sum_{n \geq 1} \frac{f(n)}{n^c}$ 绝对收敛, 那 $\lim_{n \rightarrow \infty} \frac{|f(n)|}{n^c} = 0$, 则 $f(n) \ll n^c$ 。

Exercise 2. (a) 找例子 s.t. $\sigma_a = \sigma_c + 1$; (b) 任给实数 σ_a, σ_c , $\sigma_a - 1 \leq \sigma_c \leq \sigma_a$, 求 f 使 L_f 对应参数为 σ_a, σ_c 。

Solution. 对 (a) $f(n) = (-1)^n$ 有 $\sigma_c = 0, \sigma_a = 1$; (b) $f(n) = (-1)^n n^{\sigma_a-1} + 1_{n=2^k} n^{\sigma_c}$, 前一个对应 L 级数参数为 $\sigma_a - 1, \sigma_a$, 后一个为 σ_c 。

Exercise 3. 找 $F(s) = \sum_{m_1, m_2, m_3 \geq 1} (m_1 + m_2 + m_3)^{-s}$ 的绝对收敛横坐标, 用 $\zeta(s)$ 表示。

Solution. $F(s) = \sum_{n \geq 1} \sum_{m_1+m_2+m_3=n} n^{-s} = \sum_{n \geq 3} \binom{n-1}{2} n^{-s}$, $\binom{n-1}{2} \asymp n^2/2$, 故 $\sigma_a = 3$ 。 $F(s) = \sum_{n \geq 3} \frac{n^2-3n+2}{2} n^{-s} = \frac{1}{2}(\zeta(s-2) - 3\zeta(s-1) + 2\zeta(s))$ ($n = 0, 1, 2$ 项抵消)。

Exercise 4. $\chi(n)$ 定义为 $\chi(1) = 1, \chi(p_1^{\alpha_1} \cdots p_k^{\alpha_k}) = \alpha_1 \alpha_2 \cdots \alpha_k$, 计算 (a) $\sum_{n \geq 1} \frac{\tau(n^2)}{n^s}$; (b) $\sum_{n \geq 1} \frac{2^{\omega(n)}}{n^s}$; (c) $\sum_{n \geq 1} \frac{(-1)^{\Omega(n)}}{n^s}$; (d) $\sum_{n \geq 1} \frac{3^{\omega(n)} \chi(n)}{n^s}$ 。

Solution. pf: (a) $\tau(n^2)$ 可乘, 用 Euler 乘积, $\sum_{n \geq 1} \frac{\tau(n^2)}{n^s} = \prod_p (1 + \sum_{\nu \geq 1} \frac{\tau(p^{2\nu})}{p^{\nu s}}) = \prod_p (1 + \sum_{\nu \geq 1} \frac{2\nu+1}{p^{\nu s}}) = \prod_p ((1-p^{-s})^{-1} (1-p^{-s})^{-1} (1+p^{-s})) = \prod_p (1-p^{-s})^{-3} (1-p^{-2s}) = \frac{\zeta(s)^3}{\zeta(2s)}$ 。(b) $2^{\omega(n)}$ 可乘, 原式 $= \prod_p (1 + \sum_{\nu \geq 1} \frac{2}{p^{\nu s}}) = \prod_p (1-p^{-2s}) \prod_p (1-p^{-s})^{-2} = \frac{\zeta(s)^2}{\zeta(2s)}$ 。

(c) 显然 $= \frac{\zeta(2s)}{\zeta(s)}$ (是 Liouville 函数)。 (d) 可乘, $= \prod_p (1 + \sum_{\nu \geq 1} \frac{3\nu}{p^{\nu s}}) = (\text{错位相减}) = \prod_p (1 + \frac{3p^s}{(p^s-1)^2}) = \prod_p (1 - p^{-3s})(1 - p^{-s})^{-3} = \frac{\zeta(s)^3}{\zeta(3s)}$ 。

Exercise 5. 对 $c > 0$, $\int_{(c)} = \int_{c-i\infty}^{c+i\infty}$. 证明 $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} ds = 0$ if $0 < y \leq 1$, $= \log y$ if $y \geq 1$; 用包含 Λ 的和式表示 $(2\pi i)^{-1} \int_{(2)} y^s s^{-2} (-\zeta'/\zeta)(s) ds$

Solution. 模仿讲义围道积分, 模仿讲义 perron 公式可得

$$(2\pi i)^{-1} \int_{(2)} y^s s^{-2} \left(-\frac{\zeta'}{\zeta} \right) (s) ds = \sum_{n \leq y} \Lambda(n) \log(y/n).$$

Exercise 6. $\operatorname{Re}(s) > 1$, $\frac{1}{\zeta(s)} = s \int_1^\infty \frac{M(x)}{x^{s+1}} dx$, $-\frac{\zeta'(s)}{\zeta(s)} = s \int_1^\infty \frac{\psi(x)}{x^{s+1}} dx$, 其中 $M(x) = \sum_{n \leq x} \mu(n)$, $\psi(x) = \sum_{n \leq x} \Lambda(n)$.

Solution. pf: (a) $s \int_1^N \frac{M(x)}{x^{s+1}} dx = s \sum_{n=1}^N \mu(n) \int_n^N \frac{1}{x^{s+1}} dx = -[\sum_{n=1}^N \frac{\mu(n)}{N^s} - \sum_{n=1}^N \frac{\mu(n)}{n^s}]$
 (取 $N \rightarrow \infty$) $= \sum_{n=1}^\infty \frac{\mu(n)}{n^s} = \frac{1}{\zeta(s)}$ 。 (b) $s \int_1^N \frac{\psi(x)}{x^{s+1}} dx = s \sum_{n=1}^N \Lambda(n) \int_n^N \frac{1}{x^{s+1}} dx = -[\sum_{n=1}^N \frac{\Lambda(n)}{N^s} - \sum_{n=1}^N \frac{\Lambda(n)}{n^s}] \rightarrow \sum_{n=1}^\infty \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}$ 。

第五章

Exercise Sheet 5

Exercise 1. $k = 2^\alpha$, $\forall a$ 奇, $a \equiv (-1)^{r_{-1}(a)} 5^{r_0(a)} \pmod{2^\alpha}$, 对 $\alpha \geq 2$, (a) $r_{-1}(a) \equiv \frac{a-1}{2} \pmod{2}$; (b) $a \equiv a' \pmod{2^\alpha}$ iff $r_{-1}(a') \equiv r_{-1}(a) \pmod{2}$, $r_0(a') \equiv r_0(a) \pmod{2^{\alpha-2}}$.

Solution. pf: 显, 见讲义。

Exercise 2. $\varphi^*(k)$ 是模 k 本原特征个数。求证 $\varphi^*(k) = \sum_{d|k} \mu(d) \varphi(k/d) = k \prod_{p|k} (1 - \frac{2}{p}) \prod_{p^2|k} (1 - \frac{1}{p})^2$.

Solution. pf: 每个模 k 特征唯一由一个模 $d|k$ 本原特征诱导, 故 $\varphi(k) = \sum_{d|k} \varphi^*(d) \Rightarrow \varphi^*(k) = \sum_{d|k} \mu(d) \varphi(k/d)$, 第二个等式由可乘性, 只需化 $k = p^\alpha$, $\sum_{d|p^\alpha} \mu(d) \varphi(p^\alpha/d) = \varphi(p^\alpha) - \varphi(p^{\alpha-1})$

Exercise 3. χ 模 k 的 D 特征, $k' | k$, TFAE: (a) χ 由 k' 本原特征诱导; (b) 若 $(n_i, k) = 1$ 且 $n_1 \equiv n_2 \pmod{k'}$, 则 $\chi(n_1) = \chi(n_2)$; (c) $\chi(n_1) = 1$, $\forall n_1 \equiv 1 \pmod{k'}$, $(n_1, k) = 1$.

Solution. pf: 讲义 6.3.2。

Exercise 4. $k = 2^\alpha$, $\alpha \geq 3$, $\chi(n, k, m_{-1}, m_0) = (-1)^{m_{-1}r_{-1}(n)} e(\frac{m_0 r_0(n)}{2^{\alpha-2}}) ((n, 2) = 1)$, 本原 iff $(m_0, 2) = 1$.

Solution. pf: 见 6.3.6。

Exercise 5. 模 4 非主特征本原: $\chi_4(n) = (-1)^{\frac{n-1}{2}} ((n, 2) = 1)$ 。模 8 本原实特征只有, $\chi_8(n) = (-1)^{\frac{n^2-1}{8}} ((n, 2) = 1)$, $\chi_4(n) \chi_8(n) = (-1)^{\frac{(n-1)(n+5)}{8}} ((n, 2) = 1)$ 。

Solution. pf: 直接计算。

Exercise 6. $f: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$, $\hat{f}: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$, $\hat{f}(\alpha) := \frac{1}{\sqrt{q}} \sum_{a \bmod q} f(a) e^{2\pi i \alpha a/q}$ 。(a) $f, g: \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$, 证明 $\sum_{a \bmod q} f(a) \overline{g(a)} = \sum_{\alpha \bmod q} \hat{f}(\alpha) \overline{\hat{g}(\alpha)}$; (b) $\sum_{a \bmod q} |f(a)|^2 = \sum_{\alpha \bmod q} |\hat{f}(\alpha)|^2$; (c) $I = [A, B] \subset [0, q]$, 计算 $\hat{\mathbf{1}}_I(\alpha)$; (d) 求证 $\sum_{\alpha \bmod q} |\hat{\mathbf{1}}_I(\alpha)| \ll \sqrt{q} \log q$; (e) 证明对本原 D-特征 $\chi \bmod q$, $\forall N \geq 1$, $\sum_{n=1}^N \chi(n) \ll \sqrt{q} \log q$ 。

Solution. (a) pf: 换序求和, 显, (用加法特征正交性。)(b) pf: (a) 中取 $f = g$ 。(c) pf: $\hat{\mathbf{1}}_I(\alpha) = \frac{1}{\sqrt{q}} \sum_{A \leq a \leq B} e^{2\pi i \alpha a/q}$ (等比级数)。不妨设 $A, B \in \mathbb{Z}$, (否则考虑 $[A] + 1, [B]$); $\hat{\mathbf{1}}_I(0) = \frac{1}{\sqrt{q}} (B - A + 1)$, $\hat{\mathbf{1}}_I(\alpha) = \frac{1}{\sqrt{q}} \frac{e^{2\pi i \alpha A/q} (1 - e^{2\pi i \alpha (B - A + 1)/q})}{1 - e^{2\pi i \alpha/q}}$ 。(d) pf: 其中 $L = \hat{B} - \hat{A} + 1$, LHS = $\frac{1}{\sqrt{q}} \sum_{\alpha \bmod q} \frac{|1 - e^{2\pi i \alpha L/q}|}{|1 - e^{2\pi i \alpha/q}|} \leq \sum_{\alpha} \frac{1}{\sqrt{q}} \frac{2}{|1 - e^{2\pi i \alpha/q}|} = \frac{1}{\sqrt{q}} \sum_{\alpha \bmod q} \frac{1}{|\sin(\pi \alpha/q)|} \ll \frac{2}{\sqrt{q}} \sum_{\alpha=1}^{(q+1)/2} \frac{\pi}{2\alpha\pi/q} \ll \sqrt{q} \log q$, (利用 $x \in [0, \pi/2]$, $\sin x > \frac{2}{\pi}x$ 和 $\sin x = \sin(\pi - x)$ 。)(e) pf: 见讲义 thm 6.4.9。

Exercise 7. p 素数, $(\frac{n}{p})$ 是 D 特征模 p , 求证模 p 的最小非二次剩余大小为 $O(\sqrt{p} \log p)$ 。

Solution. pf: $\chi(n) := (\frac{n}{p})$ 不是模 1 本原特征, 从而是模 p 本原特征。若 N 是最小的模 p 非二次剩余, 则 $N - 1 = \sum_{n=1}^{N-1} \chi(n) \ll \sqrt{p} \log p$, 故 $N \ll \sqrt{p} \log p$ 。

第六章

Exercise Sheet 6

Exercise 1. 有限 Abel 群 G : $G \simeq \hat{G}$.

Solution. pf: 见讲义 6.1 节。

Exercise 2. (a) $\operatorname{Re}(s) > 1$, $\zeta(s) = \frac{1}{s-1} + 1 - s \int_1^\infty \frac{\{x\}}{x^{s+1}} dx$ (给出 $\operatorname{Re}(s) > 0$ 的亚纯延拓); (b) 洛朗展开 $\zeta(s) = \frac{1}{s-1} + \gamma + O(|s-1|)$ ($s \rightarrow 1$), $\gamma = \lim_{n \rightarrow \infty} (\sum_{k=1}^n \frac{1}{k} - \log n)$.

Solution. pf: (a) Euler-Maclaurin (讲义 2.2.3), $\zeta(s) = \lim_{N \rightarrow \infty} (1 + \sum_{1 < n \leq N} \frac{1}{n^s}) = 1 + \int_1^\infty \frac{1}{x^s} dx - \int_1^\infty (\{x\} - \frac{1}{2}) \frac{x^{-s}}{x^{s+1}} dx + \lim_{N \rightarrow \infty} \frac{\{N\}-1/2}{N^s} - \frac{\{1\}-1/2}{1^s} = 1 + \frac{1}{s-1} - s \int_1^\infty \frac{\{x\}}{x^{s+1}} dx$. (b) $\zeta(s) = \frac{1}{s-1} + (1 - \int_1^\infty \frac{\{x\}}{x^2} dx) + (\int_1^\infty \frac{\{x\}}{x^2} dx - s \int_1^\infty \frac{\{x\}}{x^{s+1}} dx)$, 由讲义 2.2.4, $\gamma = 1 - \int_1^\infty \frac{\{x\}}{x^2} dx$, 而且 $\int_1^\infty \frac{\{x\}}{x^2} dx - s \int_1^\infty \frac{\{x\}}{x^{s+1}} dx = \int_1^\infty (\frac{\{x\}}{x^2} - \frac{\{x\}}{x^{s+1}}) dx - (s-1) \int_1^\infty \frac{\{x\}}{x^{s+1}} dx = \int_1^\infty \frac{\{x\}}{x^{s+1}} (\xi^{s-1} - 1) d\xi + O(s-1) = \int_1^\infty O(\frac{1}{\xi^{1+\delta}}) O((s-1) \log \xi) d\xi = O(s-1)$ ($s \rightarrow 1$).

Exercise 3. $\prod_{p \leq x} (1 - \frac{1}{p})^{-1} = C \log x + O(1)$, 求证 $C = e^\gamma$.

Solution. pf: $P(x) = \prod_{p \leq x} (1 - \frac{1}{p})^{-1}$, $L = \log x$. Ex 6-2 (b) $\Rightarrow \zeta(1 + \frac{1}{L}) = L + O(1)$, 而且 $\log \zeta(1 + \frac{1}{L}) = \sum_p -\log(1 - p^{-1-1/L}) = \sum_p \sum_{k \geq 1} \frac{p^{-k(1+1/L)}}{k} = \sum_p -\log(1 - \frac{1}{p}) p^{-1/L} + o(1)$ ($x \rightarrow \infty$). 由 Abel, $\sum_p -\log(1 - \frac{1}{p}) p^{-1/L} = \int_{1-0}^\infty y^{-1/L} d \sum_{2 \leq p \leq y} -\log(1 - \frac{1}{p}) = \frac{1}{L} \int_2^\infty y^{-1/L-1} \sum_{2 \leq p \leq y} -\log(1 - \frac{1}{p}) dy$, 故 $\log L + o(1) = \log \zeta(1 + \frac{1}{L}) = \frac{1}{L} \int_2^\infty y^{-1/L-1} \log \prod_{p \leq y} (1 - \frac{1}{p})^{-1} dy + o(1)$. 而 $C \log y + O(1) = C \log y (1 + O(\frac{1}{\log y}))$, 故上式 $= \frac{1}{L} \int_2^\infty y^{-1/L-1} [\log C + \log \log y + O(\frac{1}{\log y})] dy + o(1) = \log L + \log C - \gamma + o(1)$ (换元 $\frac{dy}{y} = d(\log y)$), 所以 $\log C = \gamma$, $C = e^\gamma$.

Exercise 4. $\chi \bmod q$, $\chi \neq \chi_0$, $x \geq 2$, 证明 (a) $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = -L'(1, \chi) + O_\chi(\frac{\log x}{x})$; (b) $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = L(1, \chi) \sum_{m \leq x} \frac{\chi(m) \Lambda(m)}{m} + O_\chi(\frac{1}{x} \sum_{n \leq x} \Lambda(n))$, 从而

$\sum_{n \leq x} \frac{\chi(n)\Lambda(n)}{n} = O_\chi(1)$; (c) $\sum_{p \leq x} \frac{\chi(p) \log p}{p} = O_\chi(1)$; (d) $\sum_{p \leq x} \frac{\chi(p)}{p} = b(\chi) + O_\chi(\frac{1}{\log x})$,
其中 $b(\chi) = \log L(1, \chi) - \sum_p \sum_{k \geq 2} \frac{\chi(p^k)}{kp^k}$.

Solution. pf: (a) $\sum_{n=1}^N \frac{\chi(n)}{n^s} = \frac{S(N)}{N^s} + s \int_1^N \frac{S(y)}{y^{s+1}} dy$, $S(y) = \sum_{n \leq y} \chi(n)$ 有界, 故 L_χ 的 $\sigma_c \leq 0$. 讲义 4.10 $\Rightarrow -L'(1, \chi) = \sum_{n \geq 1} \frac{\chi(n) \log n}{n} = \sum_{n \leq x} \frac{\chi(n) \log n}{n} + \sum_{n > x} \frac{\chi(n) \log n}{n}$,
而 $\sum_{n > x} \frac{\log n}{n} \chi(n) \stackrel{\text{Abel}}{=} \lim_{N \rightarrow \infty} [\sum_{x < n \leq N} \chi(n) \frac{\log N}{N} - \int_x^N \sum_{x < n \leq \xi} \chi(n) (\frac{\log \xi}{\xi})' d\xi] =$
 $O_\chi(\frac{\log x}{x})$. (b) $\log = \Lambda * 1$, $\sum_{n \leq x} \frac{\chi(n) \log n}{n} = \sum_{n \leq x} \sum_{d|n} \frac{\chi(d)\Lambda(d)}{d} \frac{\chi(n/d)}{n/d} =$
 $\sum_{d \leq x} \frac{\chi(d)\Lambda(d)}{d} \sum_{n \leq x/d} \frac{\chi(n)}{n} = L(1, \chi) \sum_{d \leq x} \frac{\chi(d)\Lambda(d)}{d} - \sum_{d \leq x} \frac{\chi(d)\Lambda(d)}{d} \sum_{m > x/d} \frac{\chi(m)}{m}$,
且 $\sum_{d \leq x} \frac{\chi(d)\Lambda(d)}{d} \sum_{m > x/d} \frac{\chi(m)}{m} = (dm = e) \sum_{d \leq x} \Lambda(d) \sum_{d|e} \frac{\chi(e)}{e}$, Abel 求和
 $\Rightarrow O_\chi(\frac{1}{x} \sum_{n \leq x} \Lambda(n))$. (c) $\sum_{p \leq x} \frac{\chi(p) \log p}{p} = \sum_{n \leq x} \frac{\chi(n)\Lambda(n)}{n} - \sum_{\substack{p^\ell \leq x \\ \ell \geq 2}} \frac{\chi(p)^\ell \log p}{p^\ell}$,
由 (b) 第一项有界, 第二项 $\leq \sum_{p \leq \sqrt{x}} \log p \sum_{\ell \geq 2} \frac{1}{p^\ell} = \sum_{p \leq \sqrt{x}} \frac{\log p}{p(p-1)} =$
 $O(\sum \frac{1}{n^{1.5}})$ 有界. (d) $\sum_p \sum_{k \geq 1} \frac{\chi(p^k)}{kp^k} = -\sum_p \log(1 - \frac{\chi(p)}{p}) = \log L(1, \chi)$, 故
 $\sum_p \frac{\chi(p)}{p} = \log L(1, \chi) - \sum_p \sum_{k \geq 2} \frac{\chi(p^k)}{kp^k}$. 而 $\sum_{p > x} \frac{\chi(p)}{p} = \sum_{p > x} \frac{\chi(p) \log p}{p} \frac{1}{\log p} \stackrel{\text{Abel}}{=} \sum_{x < p \leq N} \frac{\chi(p) \log p}{p} \frac{1}{\log N} - \int_x^N (\frac{1}{\log \xi})' \sum_{x < p \leq \xi} \frac{\chi(p) \log p}{p} d\xi = O_\chi(\frac{1}{\log x})$, (由 (c))

Exercise 5. $|\arg s| \leq \pi - \delta$, 求证 $\Gamma'(s)/\Gamma(s) = \log s - \frac{1}{2s} + O_\delta(|s|^{-2})$.

Solution. pf: $F(s) \triangleq \log \Gamma(s) - (s - \frac{1}{2}) \log s + s - \frac{\log(2\pi)}{2}$. 由 Stirling, $F(s) = O_\delta(|s|^{-1})$
($s \rightarrow \infty$, $|\arg s| \leq \pi - \frac{\delta}{2}$). $|s|$ 充分大时, $\exists C_\delta > 0$, $D(s, C_\delta |s|) \subset \{|\arg z| \leq \pi - \frac{\delta}{2}\}$,
故 Cauchy 积分公式 $\Rightarrow |F'(s)| \leq \frac{1}{C_\delta |s|} \sup_{|z-s|=C_\delta |s|} |F(z)| = O_\delta(\frac{1}{|s|^2})$. 故 $F'(s) =$
 $O_\delta(\frac{1}{|s|^2})$, 即为所求.

第七章

Exercise Sheet 7

引理. f 整函数, 非多项式, $f(z) = \sum_{n=0}^{\infty} a_n z^n$, 则 f 阶为 $\rho(f) = \limsup_{n \rightarrow \infty} \frac{n \log n}{-\log |a_n|}$ ($a_n = 0$ 时对应项视为 0)。

Solution. pf: Cauchy 积分公式 $\Rightarrow |a_n| \leq \sup_{|z|=r} |f(z)| \frac{1}{r^n}$ 。若 $\sup_{|z|=r} |f(z)| \ll \exp(r^{\rho+\epsilon})$, 取 $r \asymp n^{1/(\rho+\epsilon)}$, 则 $-\log |a_n| \gg \frac{n}{\rho+\epsilon} \log n + O(n)$, 故 $\limsup_{n \rightarrow \infty} \frac{n \log n}{-\log |a_n|} \leq \rho + \epsilon$ 。另一方面, 若 $-\log |a_n| \gg \frac{n \log n}{\mu}$, 则 $|a_n| r^n \leq \exp(n \log r - \frac{n \log n}{\mu})$, $n \sim r^\mu \Rightarrow \sup_{|z|=r} f(z) \leq \exp(Cr^\mu)$ (r 充分大), $\Rightarrow \rho(f) \leq \mu$, $\mu \rightarrow \limsup_{n \rightarrow \infty} \frac{n \log n}{-\log |a_n|}$ 。

Exercise 1. $\forall \alpha > 0$, $f_\alpha(z) = \sum_{j=1}^{\infty} \frac{z^j}{\Gamma(1+j/\alpha)}$ 为 α 阶整函数。

Solution. pf: $a_j \triangleq \Gamma(1+j/\alpha)^{-1}$, Stirling $\Rightarrow -\frac{1}{j} \log |a_j| = \frac{1}{\alpha} (\log j + O(1)) \rightarrow +\infty$, 故收敛半径 $R = \frac{1}{\limsup_{j \rightarrow \infty} |a_j|^{1/j}} = +\infty$, 故整。 $\limsup_{n \rightarrow \infty} \frac{n \log n}{-\log |a_n|} = \alpha$, 故阶为 α 。

Exercise 2. $f(z) \triangleq \sum_{n=0}^{\infty} e^{-n^2} z^n$ 是非多项式 0 阶整函数。

Solution. pf: 因为 $\limsup_{n \rightarrow \infty} \frac{n \log n}{-\log |e^{-n^2}|} = 0$ 。

Exercise 3. 整函数 f 阶为 α , $f(0) \neq 0$, $\forall \epsilon > 0$, $\sum_{\rho, f(\rho)=0} \frac{1}{|\rho|^{\alpha+\epsilon}} < \infty$ (计重数)。

Solution. pf: Thm 10.2.6 说 $\sum_{|\rho| < R} 1 \ll R^{\alpha+\delta}$ 。设零点最小模长为 r , 则 $\sum_{\rho} \frac{1}{|\rho|^{\alpha+\epsilon}} = \sum_{n > r} \sum_{2^n < |\rho| \leq 2^{n+1}} \frac{1}{|\rho|^{\alpha+\epsilon}} \ll \sum_{n > r} \frac{\sum_{|\rho| \leq 2^{n+1}} 1}{2^{n(\alpha+\epsilon)}} \leq \sum_{n > r} \frac{2^{(n+1)(\alpha+\delta)}}{2^{n(\alpha+\epsilon)}} = \sum_{n > r} 2^{\alpha+\delta-n(\epsilon-\delta)} < \infty$ (等比级数, 一开始取 $\delta < \epsilon$)。

Exercise 4. Hadamard $\zeta^*(s) = e^{Bs} \prod_{\rho} (1 - \frac{s}{\rho}) e^{s/\rho}$ 中 $B = -\sum_{\rho} \frac{1}{\rho} = -2 \sum_{\gamma > 0} \frac{\beta}{\beta^2 + \gamma^2}$, 其中 $\rho = \beta + i\gamma$, $\sum_{\rho} \frac{1}{\rho} \triangleq \lim_{x \rightarrow \infty} \sum_{|\rho| < x} \frac{1}{\rho}$ 。

Solution. pf: $\frac{\zeta^{*'}(s)}{\zeta^{*}(s)} = B + \sum_{\rho} (\frac{1}{s-\rho} + \frac{1}{\rho})$. $s \rightarrow 0 \Rightarrow \frac{\zeta^{*'}}{\zeta^{*}}(0) = B$. 由 $\zeta^{*}(s) = \zeta^{*}(1-s)$, $\frac{\zeta^{*'}}{\zeta^{*}}(0) = -\frac{\zeta^{*'}}{\zeta^{*}}(1) = -[B + \sum_{\rho} (\frac{1}{1-\rho} + \frac{1}{\rho})]$. 由 ρ 零点, $1-\rho$ 也为 ζ^{*} 零点, 故对称主
值下 $\sum_{\rho} \frac{1}{1-\rho} = \sum_{\rho} \frac{1}{\rho}$, 得 $B = -\sum_{\rho} \frac{1}{\rho}$. 又 $\rho = \beta + i\gamma$ 是零点时 $\bar{\rho} = \beta - i\gamma$ 也是零点,
故 $B = -\sum_{\gamma>0} (\frac{1}{\beta+i\gamma} + \frac{1}{\beta-i\gamma}) = -2\sum_{\gamma>0} \frac{\beta}{\beta^2+\gamma^2}$.

Exercise 5. 设 χ 是模 q 的本原 Dirichlet 特征, $\chi(-1) = 1$. 证明 $L(s, \chi) = \sum_{n \geq 1} \chi(n)n^{-s}$ 解析延拓到整个复平面, 并满足 $\pi^{-(1-s)/2} q^{(1-s)/2} \Gamma(\frac{1-s}{2}) L(1-s, \bar{\chi}) = \frac{\sqrt{q}}{\tau(\chi)} \pi^{-s/2} q^{s/2} \Gamma(\frac{s}{2}) L(s, \chi)$, 其中 $\tau(\chi) := \sum_{a=1}^q \chi(a) e^{2\pi i a/q}$; 并说明 $\chi(-1) = -1$ 时哪里出问题.

Solution. pf: 除 $q=1$ 的主特征外, 本原特征非主, 以下看非主情形. 定义

$$\theta(x, \chi) := \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 x/q}.$$

由 $\chi(-1) = 1$, 有 $\theta(x, \chi) = 2 \sum_{n \geq 1} \chi(n) e^{-\pi n^2 x/q}$. 当 $\operatorname{Re} s > 1$ 时逐项积分, 且

$$\int_0^{\infty} e^{-\pi n^2 x/q} x^{s/2-1} dx = (q/\pi)^{s/2} \Gamma(\frac{s}{2}) n^{-s}.$$

故记 $I(s, \chi) := \int_0^{\infty} \theta(x, \chi) x^{s/2} \frac{dx}{x}$, 则

$$I(s, \chi) = 2\pi^{-s/2} q^{s/2} \Gamma(\frac{s}{2}) L(s, \chi). \quad (*)$$

接着按 $n = a + mq$ 分解并对内层用 Poisson 求和:

$$\theta(x, \chi) = \sum_{a \bmod q} \chi(a) \sum_{m \in \mathbb{Z}} e^{-\pi(a+mq)^2 x/q},$$

而

$$\sum_{m \in \mathbb{Z}} e^{-\pi(a+mq)^2 x/q} = \frac{1}{q} \sqrt{\frac{q}{x}} \sum_{k \in \mathbb{Z}} e^{2\pi i k a/q} e^{-\pi k^2/(qx)}.$$

又 χ 本原给 $\sum_{a \bmod q} \chi(a) e^{2\pi i k a/q} = \bar{\chi}(k) \tau(\chi)$, 于是

$$\theta(x, \chi) = \frac{\tau(\chi)}{\sqrt{qx}} \theta(\frac{1}{x}, \bar{\chi}).$$

由定义知 $x \rightarrow \infty$ 时 $\theta(x, \chi)$ 指数衰减, 由变换式知 $x \rightarrow 0^+$ 时也足够衰减, 故 $I(s, \chi)$ 全纯; 再由 (*) 定义 $L(s, \chi) := \frac{1}{2} \pi^{s/2} q^{-s/2} I(s, \chi) / \Gamma(\frac{s}{2})$, 得到解析延拓. 对 $I(1-s, \bar{\chi})$ 作 $x = 1/y$, 并用上面对 $\bar{\chi}$ 的变换式, 得 $I(1-s, \bar{\chi}) = \frac{\sqrt{q}}{\tau(\chi)} I(s, \chi)$; 把 (*) 分别用于 (s, χ) 和 $(1-s, \bar{\chi})$ 即得函数方程. 若 $\chi(-1) = -1$, 则 $\theta(x, \chi)$ 中 n 与 $-n$ 两项抵消, 所以

$\theta(x, \chi) \equiv 0$, 上面的积分公式不能恢复 $L(s, \chi)$; 应改用 $\theta_1(x, \chi) = \sum_{n \in \mathbb{Z}} n \chi(n) e^{-\pi n^2 x/q}$, Gamma 因子改为 $\Gamma(\frac{s+1}{2})$ 。

第八章

Exercise Sheet 8

Exercise 1. 若存在 $C > 0$ 使 $\zeta(s)$ 任一零点 ρ 满足 $\operatorname{Re} \rho < 1 - C/\log(2 + |\operatorname{Im} \rho|)$, 不用显式公式证明存在 $c > 0$ 使 $\psi(x) = \sum_{n \leq x} \Lambda(n) = x + O(xe^{-c\sqrt{\log x}})$.

Solution. pf: 只需看非整数 x , 整数点由 Perron 半权公式或 $x \pm 0$ 逼近. 记 $L = \log x$, $L_T = \log(T + 2)$, 取 $a = 1 + 1/L$, $b = 1 - C/(2 \log T)$, $3 \leq T \leq x$. 若 $|\gamma| \leq T$, 则零点 $\rho = \beta + i\gamma$ 满足 $\beta < 1 - C/\log(2 + |\gamma|) \leq 1 - C/\log(2 + T) < b$ (小 T 吸入常数), 故矩形 $b \leq \operatorname{Re} s \leq a$, $|\operatorname{Im} s| \leq T$ 中 $-\zeta'/\zeta$ 只有 $s = 1$ 的简单极点, 且 $-\zeta'/\zeta(s) = 1/(s - 1) + O(1)$. Perron 公式给 $\psi(x) = \frac{1}{2\pi i} \int_{a-iT}^{a+iT} (-\zeta'/\zeta(s)) x^s s^{-1} ds + O(xL^2/T)$. 由柯西留数定理移线到 $\operatorname{Re} s = b$, 穿过 $s = 1$ 的留数为 x . 无零矩形中标准 Jensen–Cauchy 估计给 $-\zeta'/\zeta(s) \ll_C L_T^2$, 故新竖线为 $O_C(x^b L_T^3)$, 横边与截断误差为 $O_C(xL^2/T)$. 于是 $\psi(x) = x + O_C(x^b L_T^3 + xL^2/T)$. 取 $T = \exp(A\sqrt{L})$, $A = \sqrt{C/2}$, 则 $x^b = x \exp(-CL/(2 \log T)) = xe^{-A\sqrt{L}}$ 且 $x/T = xe^{-A\sqrt{L}}$; 吸收对数幂得 $\psi(x) = x + O(xe^{-c_1\sqrt{\log x}})$.

Exercise 2. 设 μ 为 Mobius 函数. 用 Perron 公式证明存在 $c > 0$ 使 $\sum_{n \leq x} \mu(n) = O(xe^{-c\sqrt{\log x}})$. 可用事实: 若 $\zeta(\sigma + i\tau) \neq 0$ 对 $\sigma \geq 1 - c_0/\log(|\tau| + 2)$ 成立, 则当 $\sigma \geq 1 - \frac{c_0}{2 \log(|\tau| + 2)}$ 时 $1/|\zeta(\sigma + i\tau)| \ll_{c_0} \log(|\tau| + 2)$.

Solution. pf: 设无零区域为 $\zeta(\sigma + i\tau) \neq 0$, $\sigma \geq 1 - \kappa/\log(|\tau| + 2)$. 由题中 fact, 若 $\sigma \geq 1 - \kappa/(2 \log(|\tau| + 2))$, 则 $1/|\zeta(\sigma + i\tau)| \ll_{\kappa} \log(|\tau| + 2)$. 记 $M(x) = \sum_{n \leq x} \mu(n)$, $L = \log x$, $L_T = \log(T + 2)$, 取 $a = 1 + 1/L$, $b = 1 - \kappa/(2 \log T)$. Perron 公式给 $M(x) = \frac{1}{2\pi i} \int_{a-iT}^{a+iT} x^s (\zeta(s))^{-1} ds + O(xL/T)$. 移线到 $\operatorname{Re} s = b$; 因 $1/\zeta(s)$ 在矩形内全纯且 $s = 1$ 是 $1/\zeta$ 的零点而非极点, 所以无留数. 由 fact, 竖线为 $O_{\kappa}(x^b L_T^2)$, 横边与截

断误差为 $O_\kappa(xL/T)$, 故 $M(x) = O_\kappa(x^b L_T^2 + xL/T)$. 取 $T = \exp(B\sqrt{L})$, $B = \sqrt{\kappa/2}$, 则 $x^b = xe^{-B\sqrt{L}}$ 且 $x/T = xe^{-B\sqrt{L}}$, 吸收对数幂得 $\sum_{n \leq x} \mu(n) = O(xe^{-c_2\sqrt{\log x}})$.

Exercise 3. $N \in \mathbb{N}$, $\text{Li}(x) = \frac{x}{\log x} \sum_{n=0}^{N-1} \frac{n!}{(\log x)^n} + O_N(\frac{x}{(\log x)^{N+1}})$.

Solution. pf: $\text{Li}(x) = \int_2^x \frac{1}{\log u} du$, 反复分部积分得 $\frac{x}{\log x} \sum_{n=0}^{N-1} \frac{n!}{(\log x)^n} + \int_2^x \frac{du}{(\log u)^{N+1}}$. 余项 $= \int_2^{\sqrt{x}} \frac{du}{(\log u)^{N+1}} + \int_{\sqrt{x}}^x \frac{du}{(\log u)^{N+1}} \leq \frac{\sqrt{x}}{(\log 2)^{N+1}} + \frac{x-\sqrt{x}}{(\frac{1}{2}\log x)^{N+1}} \ll_N \frac{x}{(\log x)^{N+1}}$.

Exercise 4. 记 $M(x) := \sum_{n \leq x} \mu(n)$ 与 $\psi(x) := \sum_{n \leq x} \Lambda(n)$. 则在 $0 \leq \theta \leq 1$ 下, $M(x) = O_\varepsilon(x^{1-\theta+\varepsilon})$ 当且仅当 $\psi(x) = x + O_\varepsilon(x^{1-\theta+\varepsilon})$.

Solution. pf: $\theta = 0$ 平凡, 设 $0 < \theta \leq 1$, 置 $\alpha = 1 - \theta < 1$. 先证两边都等价于 $\zeta(s) \neq 0$ 对 $\text{Re } s > \alpha$. 若 $M(x) = O_\varepsilon(x^{\alpha+\varepsilon})$, 则 $\text{Re } s > 1$ 时 $1/\zeta(s) = \sum \mu(n)n^{-s} = s \int_1^\infty M(t)t^{-s-1} dt$; 对 $\sigma = \text{Re } s > \alpha$ 取 $\eta > 0$ 使 $\alpha + \eta < \sigma$, 被积函数为 $O(t^{\alpha+\eta-\sigma-1})$, 故积分在 $\text{Re } s > \alpha$ 局部一致收敛, 给出 $1/\zeta$ 全纯延拓, 于是 ζ 无零. 若 $\psi(x) = x + O_\varepsilon(x^{\alpha+\varepsilon})$, 写 $\psi = x + E$, 则 $\text{Re } s > 1$ 时 $-\zeta'/\zeta(s) = s \int_1^\infty \psi(t)t^{-s-1} dt = \frac{s}{s-1} + s \int_1^\infty E(t)t^{-s-1} dt$, 同理后一积分在 $\text{Re } s > \alpha$ 全纯, 故 $-\zeta'/\zeta(s) - 1/(s-1)$ 全纯; 若 ζ 在此半平面有零点, 则 $-\zeta'/\zeta$ 有极点, 矛盾. 反过来设 $\zeta(s) \neq 0$ 对 $\text{Re } s > \alpha$. 任取 $\varepsilon > 0$, 取 $b = \alpha + \varepsilon/3$, $c = 1 + 1/\log x$, $T = x$. 在 $\text{Re } s \geq b$ 中用标准零点避开估计 $1/\zeta(s) \ll_\varepsilon (|t|+2)^{\varepsilon/3}$ 与 $-\zeta'/\zeta(s) \ll_\varepsilon \log^2(|t|+3)$. Perron 公式给 $M(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{x^s}{s\zeta(s)} ds + O(\log x)$, 移线到 $\text{Re } s = b$, 不穿过留数, 新竖线 $\ll_\varepsilon x^b \int_0^T (t+2)^{\varepsilon/3}(t+1)^{-1} dt = O_\varepsilon(x^{\alpha+\varepsilon})$, 横边与截断误差同阶, 故 $M(x) = O_\varepsilon(x^{\alpha+\varepsilon})$. 同样 $\psi(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} (-\zeta'/\zeta(s))x^s s^{-1} ds + O(\log^2 x)$, 移线只穿过 $s=1$, 因 $-\zeta'/\zeta(s) = 1/(s-1) + O(1)$, 留数为 $\text{Res}_{s=1}((- \zeta'/\zeta(s))x^s/s) = x$; 余下竖线 $\ll_\varepsilon x^b \int_0^T \log^2(t+3)(t+1)^{-1} dt = O_\varepsilon(x^{\alpha+\varepsilon})$, 故 $\psi(x) = x + O_\varepsilon(x^{\alpha+\varepsilon})$. 整数 x 处用 Perron 半权公式或 $x \pm 0$ 逼近即可, 代回 $\alpha = 1 - \theta$ 得证.

命题 8.0.1 (柯西积分公式 (Cauchy integral formula)) 设 $\Omega \subset \mathbb{C}$ 为区域, f 在 Ω 上全纯. 若 $\overline{D(z_0, r)} \subset \Omega$, 则对任意 $n \geq 0$, $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{|\zeta-z_0|=r} \frac{f(\zeta)}{(\zeta-z_0)^{n+1}} d\zeta$.

留数定理 (residue theorem). 设 γ 为正向简单闭曲线, f 在 γ 及其内部除有限个孤立奇点 $a_1, \dots, a_m$ 外全纯, 则 $\int_\gamma f(z) dz = 2\pi i \sum_{j=1}^m \text{Res}(f, a_j)$.